Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

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Relational Inventory Control & Stock Tracking System

Presented By:

M.PRANOJ REDDY – VTU25893 – 23UECS0344

Courses Handling Faculty:

Dr.Hafizur Rahman(TTS3985)

INTRODUCTION

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- Inventory management using manual methods often leads to errors and data loss.
- Businesses need an efficient system to track stock levels and product details.
- This project is important to automate inventory control and improve accuracy.
- It reduces time consumption and minimizes human mistakes in stock handling.
- The system is useful in real-world applications like retail stores and warehouses.
- It allows users to add, update, delete, and search product records.
- The project uses web technologies and a database for dynamic data handling.
- Overall, it provides a simple, reliable, and user-friendly inventory tracking solution.

PROBLEM STATEMENT

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- In many small businesses, inventory management is still handled manually or with basic tools.
- This leads to errors in stock tracking, data duplication, and loss of important information.
- Manual systems are time-consuming and inefficient for handling large amounts of data.
- It becomes difficult to monitor product availability and update stock in real time.
- Lack of proper organization affects decision-making and business performance.
- There is no easy way to search, update, or maintain product records quickly.
- Therefore, there is a need for an automated and database-driven system.
- The system should be simple, accurate, and efficient for managing inventory and stock tracking.

OBJECTIVE(S)

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- To design and develop a web-based Relational Inventory Control & Stock Tracking System for efficient product management.
- To implement CRUD operations (add, update, delete, search) to improve accuracy and speed of inventory handling.

PROPOSED SYSTEM

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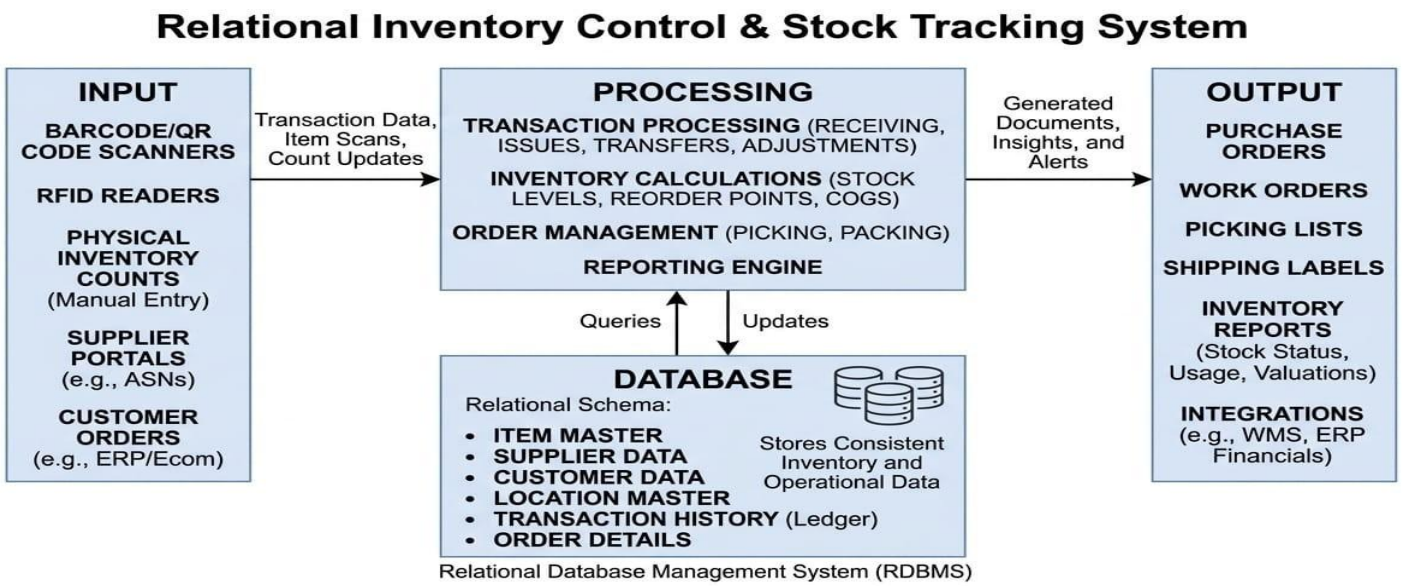
- The proposed system is a web-based inventory control and stock tracking application using PHP and MySQL.
- It provides a centralized database to store and manage product details efficiently.
- Users can perform CRUD operations such as adding, updating, deleting, and viewing products.
- A search feature allows quick retrieval of product information.
- The system reduces manual errors by automating data entry and calculations.
- It saves time and improves productivity compared to traditional manual systems.
- The user-friendly interface makes it easy to operate without technical knowledge.
- Overall, it offers a reliable, accurate, and efficient solution for real-time inventory management.

PROPOSED SYSTEM ARCHITECTURE



- **Input:**
- Users enter product details such as name, quantity, and price through web forms.
- **Processing:**
- The system processes user requests using PHP scripts (add, update, delete, search).
- **Database:**
- MySQL database (inventory_db) stores product information in the products table.
- **Output:**
- Processed data is displayed as tables (product list) on web pages.

WORKFLOW

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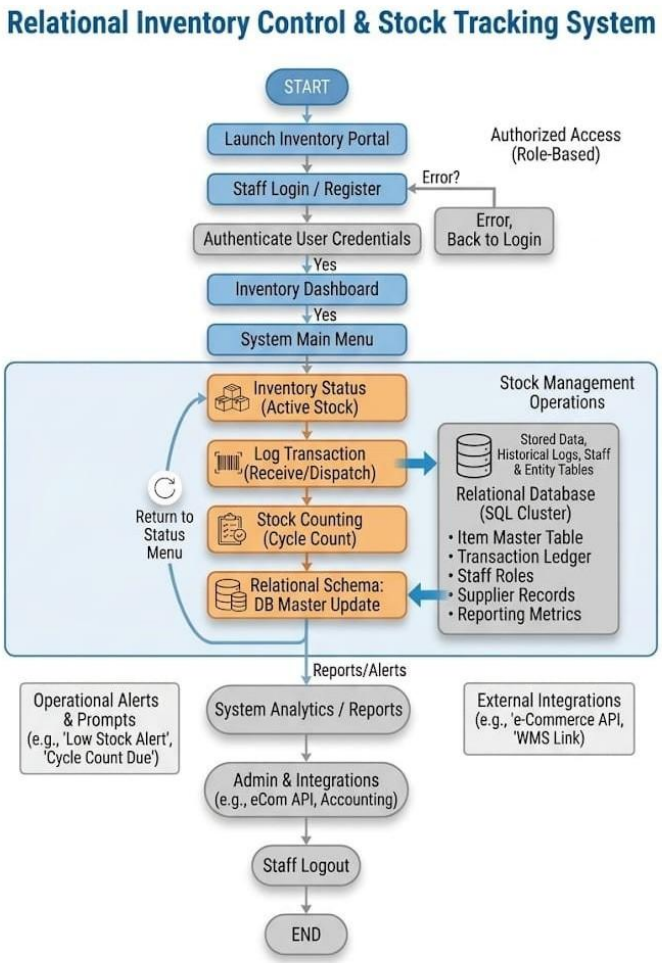

TECHNOLOGIES USED

- **Frontend:**
- HTML5,CSS3,JavaScript(ES6).
- **Backend:**
- PHP(Core PHP for authentication).
- **Database:**
- MYSQL (database,products table).
- **Tools:**
- XAMPP(Apache server+MySQL),VS Code,phpMyAdmin,Web browser.
- **Hardware/Software Requirements:**
- Minimum 4GB RAM,Windows 10/11,XAMPP Server,VS Code,Web Server.

MODULE DESCRIPTION

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Module 1: Product Management
Handles adding new products with details like name, quantity, and price.
Allows users to maintain accurate product records**.**
Module 2: Inventory Operations
Implements core functionalities such as update, delete, and stock modification.
Ensures proper tracking of available stock levels.
Module 3: Database Management
Manages storage and retrieval of product data using MySQL database.
Ensures data consistency, integrity, and efficient access.
Module 4: Search & Display
Provides search functionality to quickly find products.
Displays product list in a structured table format for easy understanding.

IMPLEMENTATION SCREENSHOTS

- Acts as the primary portal for adding new inventory items into the system's relational database.
- Captures the product identifier (SKU or Description), which links directly to the Item Master table.
- Records the physical count of the item, updating the Stock Levels in real-time.

IMPLEMENTATION SCREENSHOTS

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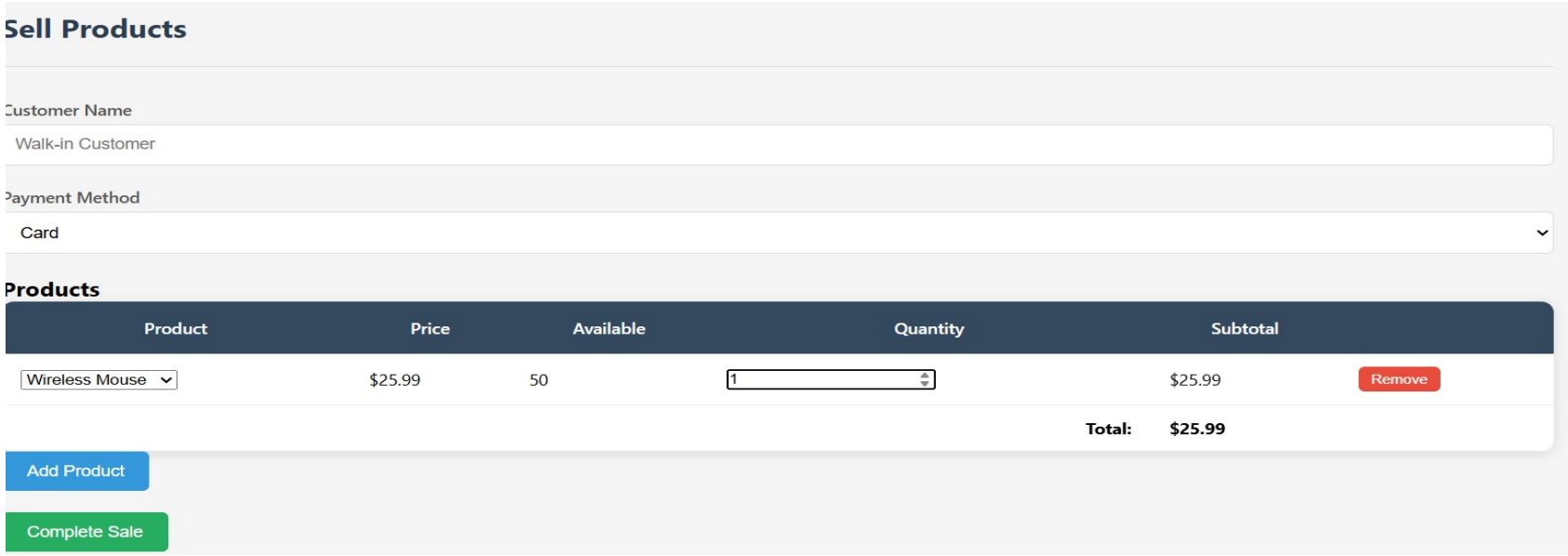
Products

Add Product

Code	Name	Category	Supplier	Price	Stock	Status
PRD005	WiFi Router	Networking	TechDistributors Inc.	$89.99	15	In Stock
PRD004	Office Chair	Furniture	Furniture Plus	$199.99	10	In Stock
PRD003	A4 Paper Box	Stationery	Office Supplies Co.	$45.00	20	In Stock
PRD002	USB-C Cable 2m	Computer Accessories	TechDistributors Inc.	$12.50	100	In Stock
PRD001	Wireless Mouse	Computer Accessories	TechDistributors Inc.	$25.99	50	In Stock

- The Primary Key from the database, ensuring each product record is unique and traceable.
- Shows current stock levels, allowing for immediate identification of low-stock items.

IMPLEMENTATION SCREENSHOTS



- This triggers a targeted SQL SELECT query using a LIKE clause to filter the database and display only relevant matches.
- By narrowing down the hundreds of potential entries to just one specific item (ID 2, Laptop), the system demonstrates its ability to handle Relational Data Retrieval efficiently.

ADVANTAGES & LIMITATIONS

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ADVANTAGES:

- Reduces manual work and minimizes human errors in inventory management.
- Provides fast and easy access to product information.
- Supports CRUD operations (Add, Update, Delete, Search) efficiently.
- Improves accuracy and consistency of data using a centralized database.
- User-friendly interface makes it easy to use for beginners.

LIMITATIONS:

- Requires basic technical knowledge to set up and maintain the system.
- Depends on local server (XAMPP), not accessible online by default.
- Limited security features (no authentication unless added).
- Not suitable for very large-scale enterprise applications.
- Requires manual data entry for all products.

Thank you